

EMCH 201: Introduction to Application of Numerical Methods

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Assignment 7

Assignment due date: **11:59 PM, Monday, December 6th, 2021**

Point Possible: 100

Submission requirements:

- 1) Detailed problem-solving process needs to be provided for full points
- 2) Upload readable electronic copy to blackboard in a single file
- 3) Late submission will not be graded

1. Solve the following ordinary differential equation over the interval from $t = 0$ to $t = 0.5$ where the initial condition is $t = 0$ and $y = 1$. When step size $h = 0.5$, please use Henu's method (100 Points).

$$\frac{dy}{dt} = 2yt^2 - y$$